General Mathematics — Exercise Sheet 16

Complex exponentials and real polynomials

Variant: Statements

Exercise 1 (An exponential equation with two terms). Find every $z \in \mathbb{C}$ satisfying $e^z + e^{-z} = 0$. Use the definition and product law of Lecture 16, without choosing a single inverse for the complex exponential.

Exercise 2 (Divisibility determines two coefficients). For $a, b \in \mathbb{R}$, let $P(z) = z^4 + az^2 + bz + 1$. Find all parameter pairs for which $z^2 + 1$ divides $P(z)$. For those pairs, give every complex root with its multiplicity.

Exercise 3 (A real quartic as its parameter varies). For every $a \in \mathbb{R}$, factor $P_a(x) = x^4 + ax^2 + 1$ into real irreducible factors. State which values of a give real roots. At $a = -2$ and $a = 2$, give all complex roots with multiplicities. Use Lecture 16's meaning of irreducible for a real quadratic.

Exercise 4 (Phasors with a bounded added amplitude). For $a, b \in \mathbb{R}$, consider the sum of two real motions at angular frequency 2:

$$x(t) = \cos 2t + 2 \sin 2t, \qquad y_{a,b}(t) = a \cos 2t + b \sin 2t.$$

Find the unconstrained parameter pairs making the total motion identically zero. Then impose $a^2 + b^2 = 1$ and find the smallest and largest possible total amplitudes and all parameters attaining them. Give a phase and the least positive period of the total motion at each extremum. Use the phasor convention of Lecture 16 and Cauchy–Schwarz from Lecture 12.

Exercise 5 (A damped complex mode with real time). Let $F(t) = e^{(-2+3\mathrm{i})t}$ for $t \in \mathbb{R}$, and let $u(t) = \operatorname{Re} F(t)$. Use the real-parameter derivative formula to verify

$$u'' + 4u' + 13u = 0, \qquad u(0) = 1, \quad u'(0) = -2.$$

Prove the bound $|u(t)| \le e^{-2t}$, find every zero with $t \ge 0$, and show that u has no positive period. Use Lecture 3's geometric-sequence limit for the last claim. The differential equation is given for verification; no general solution method is assumed.